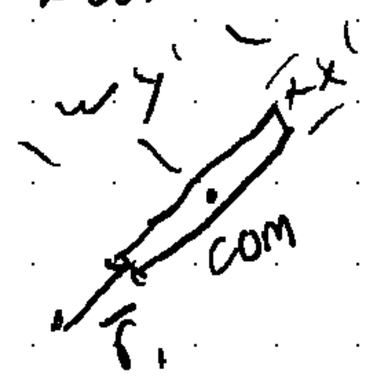
Video, Module 2: Constraints / Kinematics in MBD!

The Door!



$r_n = \bar{r}_1 + x_n \hat{i} + y_n \hat{j}$

1 ↳ $\bar{r}_{hinge} = \bar{r}' - w/2\, \hat{i}' + t/2\, \hat{j}$

2 $\bar{r}_{hinge} = \bar{r}^0 + m_x \hat{i}^0 + m_y \hat{j}^0$

door = w & t

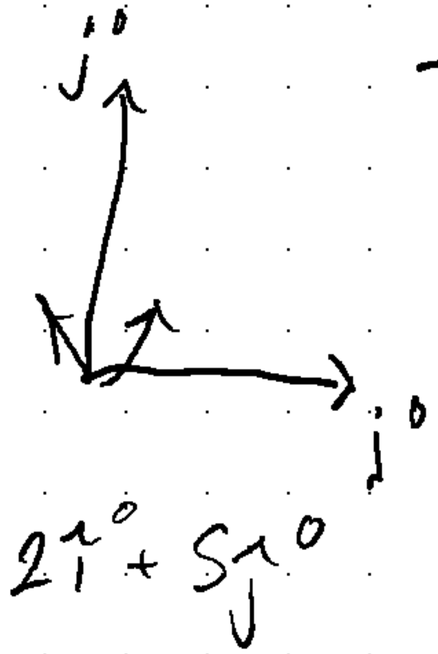
Hinge Created 2 constraint Equations $\bar{r}' - w/2\, \hat{i}' + t/2\, \hat{j}' = \bar{r}^0 + m_x \hat{i}^0 + m_y \hat{j}^0$

↳ Defined ~~point~~ By point of cm in door

↳ Defined by the ground

$$\bar{q} = \begin{bmatrix} \bar{r}' \\ \theta' \end{bmatrix} = \begin{bmatrix} r'_x \\ r'_y \\ \theta' \end{bmatrix}$$

$\bar{C}(\bar{q}, t) = \bar{0}$ ↳ $\bar{r}' - \frac{w}{2}\hat{j} + \frac{t}{2}\hat{j}' - \bar{r}^0 - m_x \hat{i}^0 - m_y \hat{j}^0 = \bar{0}$



$$\rightarrow \hat{i}^1 = \cos\theta^1\,\hat{i}^0 + \sin\theta^1\,\hat{j}^0$$

$$\hat{j} = -\sin\theta\,\hat{i}^0 + \cos\theta^1\,\hat{j}^0$$

$$\begin{matrix}\hat{i}\\ \hat{j}\end{matrix} = \begin{bmatrix}\cos\theta^1 & \sin\theta^1\\ -\sin\theta^1 & \cos\theta^1\end{bmatrix}\begin{bmatrix}\hat{i}^0\\ \hat{j}^0\end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix}x'\\ y'\end{bmatrix} = \begin{bmatrix}\cos\theta^1 & \sin\theta^1\\ -\sin\theta^1 & \cos\theta^1\end{bmatrix}\begin{bmatrix}2\\ 5\end{bmatrix}$$

Missed a few steps

$$\begin{bmatrix}c\theta^1 & -s\theta^1\\ s\theta^1 & \cos^1\end{bmatrix}\begin{bmatrix}\hat{i}^1\\ \hat{j}^1\end{bmatrix} = \begin{bmatrix}\hat{i}^0\\ \hat{j}^0\end{bmatrix}$$

$$\bar{A}\begin{bmatrix}-W/2\\ t/2\end{bmatrix}$$ global position of Hinge

So For $\bar{r}' + \bar{A}\bar{r}'_{hinge} - \bar{r}^G - \bar{A}^0\bar{r}^0_{hinge} = \bar{C}(\bar{q};t) = 2$ equations, E1

Goal is to systematically turn intuition into easy to follow set of Instructions

Turn constraints of motion into a formula I can put into any solver

End up with differential algebraic equations

Position defines velocity as well

E1 → E2 $\quad \bar{V}' + \frac{d}{dt}(\bar{A})\bar{r}' + \bar{A}\,\frac{d}{dt}(\bar{r}'_{hinge}) = 0$

General Form of velocity constraint $\frac{d\bar{C}}{d\bar{q}}\dot{\bar{q}} = -\frac{dC}{dt}$

$r_x' + -w/2\,c\theta' - t/2\,s\theta' = 0$, x & y components of

$r_y' + -w/2\,s\theta' + t/2\,c\theta' = 0$

$$\frac{d\bar{C}}{d\bar{q}}\begin{bmatrix} v_x' \\ v_y' \\ \theta' \end{bmatrix} = \begin{bmatrix} dC^1/dt \\ dC^2/dt \\ dC^3/dt \end{bmatrix}$$

$$\frac{d\bar{C}}{dq} = \left[\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \quad \bar{A}\theta' \underbrace{\begin{bmatrix} -w/2 \\ t/2 \end{bmatrix}}_{\bar{r}_{hinge}}\right]$$

$$\frac{d\bar{C}}{d\bar{q}}\begin{bmatrix} v_x' \\ v_y' \\ \theta' \end{bmatrix} = \begin{bmatrix} dC^1/dt \\ dC^2/dt \\ dC^3/dt \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & A\theta' \begin{bmatrix} -W/2 \\ +/2 \end{bmatrix} \\ 0 & 1 & \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_x' \\ v_y' \\ \theta' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$v_y' = 2$

Video Summary: $x - y = 0$ is easier for computer than $x = y$

Constraint EQs

Solve Constraint EQs